

Single-edge deletion never preserves Seidel energy in complete tripartite graphs with parts of size at least two

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Abstract

Let $G = K_{a,b,c}$, where $a, b, c \geq 2$ are integers, and let e join the parts of sizes a and b . We prove that deleting e changes the Seidel energy: $E_S(G - e) \neq E_S(G)$. Thus the equality set is empty throughout this domain. A theorem of Liu and Wang reduces the endpoint sizes to sixteen pairs. Uniform inertia formulas then express equality as an algebraic relation between one eigenvalue of a three-dimensional quotient and two eigenvalues of a five-dimensional quotient. Two quadratic factorizations and fourteen additive-compound resultants exclude the remaining families by 186 exact residue checks, using primes at most 29. These certificates cover the unbounded third part. We do not classify the sign of every energy change.

1 Statement and context

For a finite simple graph H with adjacency matrix $A(H)$, its Seidel matrix and Seidel energy are

$$S(H) = J - I - 2A(H), \quad E_S(H) = \sum_{\lambda \in \text{Spec } S(H)} |\lambda|,$$

where eigenvalues are counted with multiplicity and J is the all-ones matrix. The part containing neither endpoint of the deleted edge is marked throughout: only the two endpoint parts may be interchanged.

Theorem 1. *For every three integers $a, b, c \geq 2$ and every edge e joining the a -part and the b -part of $G = K_{a,b,c}$,*

$$E_S(G - e) \neq E_S(G).$$

Consequently the set of equality triples, modulo $(a, b, c) \sim (b, a, c)$, is empty.

Liu and Wang [1] give sufficient conditions for strict increase under edge deletion in independent-set blow-ups and a reduced spectral criterion for complete multipartite graphs. Their Theorem 3.3 and Corollary 4.1 provide the following input, with no restriction on the third part beyond its being nonempty:

Lemma 2 (Liu–Wang). *For the graphs in Theorem 1, $E_S(G - e) > E_S(G)$ if either $a, b \geq 4$, or the endpoint sizes are 3 and at least 6, or they are 2 and at least 15.*

The strict boundary $(a, b) = (4, 4)$ is included in the cited theorem; it is not inferred here from a non-strict numerical bound. After normalizing $a \leq b$, its complement consists exactly of

$$(2, b), \quad 2 \leq b \leq 14; \quad (3, b), \quad 3 \leq b \leq 5. \quad (1)$$

Every $c \geq 2$ remains to be treated for each of these sixteen pairs.

The remark following Example 4.5 of [1] leaves existence and classification of equality cases for complete multipartite graphs as a question. Theorem 1 supplies the classification for three parts of size at least two. Earlier work of Tian, Li and Cui [2] concerns tripartite Turán graphs; the introduction of [1] reports strict increase for their balanced case on at least nine vertices. Our argument retains arbitrary, independently varying part sizes. As adjacent methodological context, exact certification of graph inertia also appears in Zeng’s work on the line-graph signature inequality [3]; that citation supplies no Seidel-energy assertion used in this proof. The source comparison and its limits are recorded in Section 5.

2 Quotient spectra and uniform inertia

Put $n = a + b + c$, write the parts as A, B, C , and let $e = uv$ with $u \in A$ and $v \in B$. The original three-cell quotient, in the basis of cell indicators, is

$$Q_3 = \begin{pmatrix} a-1 & -b & -c \\ -a & b-1 & -c \\ -a & -b & c-1 \end{pmatrix}. \quad (2)$$

After deletion use the cells $\{u\}, A \setminus \{u\}, \{v\}, B \setminus \{v\}, C$. Their row sums give

$$Q_5 = \begin{pmatrix} 0 & a-1 & 1 & 1-b & -c \\ 1 & a-2 & -1 & 1-b & -c \\ 1 & 1-a & 0 & b-1 & -c \\ -1 & 1-a & 1 & b-2 & -c \\ -1 & 1-a & -1 & 1-b & c-1 \end{pmatrix}. \quad (3)$$

A vector supported on one cell and summing to zero is an eigenvector with eigenvalue -1 . The orthogonal complement of all these subspaces is the cell-constant subspace. Hence, as multisets,

$$\text{Spec } S(G) = \{-1\}^{n-3} \cup \text{Spec } Q_3, \quad \text{Spec } S(G - e) = \{-1\}^{n-5} \cup \text{Spec } Q_5. \quad (4)$$

Here $\{-1\}^k$ denotes k forced copies; a quotient can itself have additional copies of -1 . All cells are nonempty even when $a = 2$ or $b = 2$. If D is the diagonal matrix of cell sizes, then $D^{1/2} Q D^{-1/2}$ is symmetric. Both quotients are therefore diagonalizable with real spectra.

Lemma 3. *The quotient Q_3 has exactly one negative and two positive eigenvalues. The quotient Q_5 has exactly two negative and three positive eigenvalues. Neither has a zero eigenvalue for $a, b, c \geq 2$.*

Proof. In symmetric coordinates Q_3 is

$$\text{diag}(2a-1, 2b-1, 2c-1) - zz^\top, \quad z = (\sqrt{a}, \sqrt{b}, \sqrt{c})^\top.$$

It is positive definite on $z^\perp$, so has at least two positive eigenvalues. Its characteristic polynomial is

$$p_3(x) = x^3 + (3 - a - b - c)x^2 + (3 - 2a - 2b - 2c)x + 4abc - a - b - c + 1. \quad (5)$$

The constant term is 27 at $(2, 2, 2)$ and strictly increases in each variable on $[2, \infty)^3$. Thus $\det Q_3 < 0$, proving its asserted inertia.

Omit the C cell from the symmetric version of Q_5 . The resulting principal four-dimensional block is the quotient of $S(K_{a,b} - e)$. Write $N = a + b \geq 4$. Switching signs on the b -part transforms the full bipartite Seidel matrix into

$$J_N - I_N - 2(E_{uv} + E_{vu}),$$

where E_{ij} is a matrix unit. The vector $e_u - e_v$ has eigenvalue 1. Vectors supported on the other $N - 2$ vertices and summing to zero have eigenvalue -1 , with multiplicity $N - 3$. The remaining two-dimensional quotient, on vectors constant on $\{u, v\}$ and its complement, is

$$\begin{pmatrix} -1 & N-2 \\ 2 & N-3 \end{pmatrix}, \quad \det = 7 - 3N < 0.$$

The full bipartite matrix has two positive and $N - 2$ negative eigenvalues. Removing the $N - 4$ forced negative eigenvalues belonging to the four-cell partition leaves two of each sign in its quotient. Interlacing gives at least two eigenvalues of each sign in Q_5 .

Direct expansion yields

$$\det Q_5 = 12abc - 16ac - 16bc + 21c - 3a - 3b + 7.$$

Under $a = u_0 + 2$, $b = v_0 + 2$, $c = w_0 + 2$, this becomes

$$12u_0v_0w_0 + 24u_0v_0 + 8u_0w_0 + 13u_0 + 8v_0w_0 + 13v_0 + 5w_0 + 5 > 0.$$

There are no zero eigenvalues, and the number of negative eigenvalues is even. The interlacing bounds therefore force exactly two negative and three positive eigenvalues. $\square$

Let r be the negative eigenvalue of Q_3 , and let s_1, s_2 be the two negative eigenvalues of Q_5 . Since a Seidel matrix has trace zero, its energy is minus twice the sum of its negative eigenvalues. Equation (4) now gives

$$E_S(G - e) - E_S(G) = 2(r - s_1 - s_2 - 2). \quad (6)$$

In particular, the exact equality condition is

$$s_1 + s_2 = r - 2. \quad (7)$$

The inertia argument includes coincident part sizes and repeated quotient roots. It does not require any order relation between b and c .

3 The two equal-endpoint families

For $a = b = m \in \{2, 3\}$, determinant expansion factors the characteristic polynomials as

$$\begin{aligned} p_3(x) &= (x - (2m - 1))q_m(x), \\ q_m(x) &= x^2 + (2 - c)x + 1 - (2m + 1)c, \end{aligned} \quad (8)$$

$$\begin{aligned} p_5(x) &= (x^2 + (4 - 2m)x + 7 - 6m)f_m(x), \\ f_m(x) &= x^3 + (1 - c)x^2 - (2mc + 1)x + (2m - 3)c - 1. \end{aligned} \quad (9)$$

The unique negative root of q_m is r . The quadratic factor of p_5 has one negative root

$$\tau = m - 2 - \sqrt{m^2 + 2m - 3}$$

and one positive root, so Lemma 3 implies that f_m has exactly one negative root s . Put

$$d = \sqrt{m^2 + 2m - 3} - m, \quad h_m(d) = d^2 + 2md - (2m - 3) = 0.$$

Because $\tau = -2 - d$, equation (7) requires $s = r + d$. Thus equality implies $q_m(r) = f_m(r + d) = h_m(d) = 0$, and hence

$$P_m(c) := -\text{Res}_d(\text{Res}_x(q_m(x), f_m(x + d)), h_m(d)) = 0. \quad (10)$$

We use the convention $\text{Res}(f, g) = \prod_{f(\alpha)=0} g(\alpha)$ when f is monic. The resultant vanishes whenever the two polynomials have a common root. Direct exact evaluation in (10) gives

$$\begin{aligned} P_2(c) &= 125c^6 + 400c^5 - 21891c^4 + 89168c^3 \\ &\quad - 97793c^2 + 25248c - 121, \\ P_3(c) &= 147c^6 - 2898c^5 - 123967c^4 + 999156c^3 \\ &\quad - 2064123c^2 + 1248318c - 13689. \end{aligned}$$

Their values over the complete residue systems are

$$\begin{aligned} (P_2(0), P_2(1), P_2(2)) &\equiv (2, 2, 2) \pmod{3}, \\ (P_3(0), \dots, P_3(4)) &\equiv (1, 4, 3, 2, 2) \pmod{5}. \end{aligned}$$

Neither polynomial can vanish at an integer. Both families are excluded. The elimination allows both roots of h_m , which only enlarges the necessary condition. No converse implication or removal of extraneous solutions is needed because no integer candidate survives.

4 Pair sums and the other fourteen families

Let $W = Q_5^{[2]}$ be the second additive compound, acting on the exterior square of the five-dimensional space by

$$W(v \wedge w) = (Q_5 v) \wedge w + v \wedge (Q_5 w).$$

Its dimension is ten. In the lexicographic basis $e_i \wedge e_j$, $i < j$, its entries are explicitly

$$W_{(i,j),(k,l)} = (Q_5)_{ik}\delta_{jl} - (Q_5)_{il}\delta_{jk} - (Q_5)_{jk}\delta_{il} + (Q_5)_{jl}\delta_{ik}. \quad (11)$$

Diagonalizing Q_5 shows that the eigenvalues of W are all ten unordered pair sums of its eigenvalues, counted with multiplicity. Define $p_{10}(t) = \det(tI_{10} - W)$. If equality holds, then

$$p_3(r) = p_{10}(r - 2) = 0.$$

For fixed integers a, b , the following polynomial in $\mathbb{Z}[c]$ must therefore vanish:

$$R_{a,b}(c) = \text{Res}_x(p_3(x), p_{10}(x - 2)). \quad (12)$$

There is a useful determinant expression with no divisions:

$$R_{a,b}(c) = \det(Q_3 \otimes I_{10} - I_3 \otimes (W + 2I_{10})). \quad (13)$$

Both sides are the product of $\lambda - \mu - 2$ over the three eigenvalues λ of Q_3 and the ten eigenvalues μ of W . This gives an integer-polynomial identity, so it remains valid under specialization and reduction modulo any prime. Repeated eigenvalues do not affect the argument.

Lemma 4. *For every pair in (1) other than (2, 2) and (3, 3), $R_{a,b}(c) \neq 0$ for every integer c .*

Proof. Table 1 gives a prime p and all the residues $R_{a,b}(0), \dots, R_{a,b}(p-1)$ modulo p . These are obtained by integer polynomial evaluation or, equivalently, by the order-30 determinants in (13). Every entry is nonzero. An integer c is congruent to one of these representatives, so $R_{a,b}(c)$ cannot be zero. The table can be reconstructed directly from (2), (3), (11) and (13); a short exact algorithm is given in the appendix and its executable form accompanies this article. $\square$

Proof of Theorem 1. Exchange the two endpoint parts if necessary so that $a \leq b$. Lemma 2 excludes all pairs outside (1). For (2, 2) and (3, 3), the preceding quadratic elimination excludes equality for every integer $c \geq 2$. For the other fourteen pairs, equality would force (12) to vanish, contradicting Lemma 4. This exhausts the stated domain. $\square$

(a, b)	p	$R_{a,b}(0), \dots, R_{a,b}(p-1) \pmod{p}$
(2, 3)	5	2, 1, 3, 2, 4
(2, 4)	17	9, 12, 6, 11, 5, 12, 15, 4, 4, 9, 9, 14, 2, 1, 1, 12, 10
(2, 5)	13	6, 2, 7, 7, 12, 9, 9, 6, 4, 3, 5, 8, 12
(2, 6)	11	8, 3, 3, 8, 3, 5, 8, 4, 1, 2, 10
(2, 7)	13	12, 4, 11, 7, 1, 7, 2, 9, 12, 10, 4, 9, 6
(2, 8)	5	2, 1, 3, 2, 4
(2, 9)	17	11, 7, 11, 12, 13, 6, 14, 5, 15, 1, 12, 11, 8, 2, 10, 16, 16
(2, 10)	7	6, 4, 6, 1, 5, 2, 3
(2, 11)	29	7, 16, 21, 14, 6, 15, 21, 8, 6, 6, 11, 28, 28, 1, 5, 26, 14, 12, 13, 24, 3, 22, 23, 28, 4, 9, 26, 7, 2
(2, 12)	29	23, 16, 17, 22, 16, 9, 9, 9, 11, 25, 13, 16, 28, 2, 18, 26, 23, 11, 16, 26, 17, 13, 22, 23, 13, 26, 10, 27, 6
(2, 13)	5	2, 1, 3, 2, 4
(2, 14)	11	5, 6, 6, 5, 6, 2, 7, 4, 3, 4, 1
(3, 4)	5	1, 2, 2, 1, 2
(3, 5)	11	8, 1, 9, 1, 8, 9, 1, 1, 4, 2, 1

Table 1: Complete nonzero residue certificates for the fourteen unequal-endpoint families. Entries are ordered by $c = 0, \dots, p-1$; line wrapping does not restart that order.

5 Verification, provenance and limits

The proof uses exact algebra throughout. The two symmetric certificates cover $3 + 5$ residue classes, and Table 1 covers another 178, for a total of 186. This is a finite verification of complete congruence obstructions for unbounded integer families. It is not a scan of a bounded range of graph sizes. In particular, residues 0 and 1 are algebraic specializations of the certificate matrices; they need not represent graphs in the theorem’s domain.

The accompanying data include the two degree-six polynomials and all fourteen degree-fourteen resultants. A symbolic implementation constructs the additive compound by its action on basis wedges. A second implementation uses the entry formula (11) and Gaussian elimination over finite fields, without a symbolic algebra package. A separate mathematical and computational review also reconstructed the pair-sum polynomials via the ordered-sum identity

$$\text{Res}_x(f(x), f(z-x)) = 32f(z/2)p_{10}(z)^2, \quad f(x) = \det(xI_5 - Q_5). \quad (14)$$

Indeed its factors enumerate all 25 ordered sums; the five diagonal factors yield $32f(z/2)$ and each off-diagonal sum occurs twice. The review checked the full residue table and the symbolic resultant coefficients. Additional direct full-matrix checks of characteristic polynomials, inertia and rationally isolated energy differences were finite diagnostics; the universal conclusion rests on the preceding proof and certificates.

The primary dependency is the fixed version arXiv:2607.06095v1 of Liu–Wang [1], whose strictness theorem and multipartite specialization were inspected. The literature comparison also used saved arXiv searches for Seidel energy under edge deletion and for Seidel tripartite graphs, inspected on September 9, 2026. No equivalent complete classification was identified in those inspected sources. This is a bounded literature assessment, not a priority claim. The full text of [2] was unavailable in that comparison, so its uninspected subsidiary results are not ruled out as earlier equivalents. The balanced result mentioned in Section 1 is attributed through the inspected Liu–Wang introduction.

The theorem neither supplies a complete partition into strict increase and strict decrease nor addresses parts of size one or graphs with four or more parts. The broader multipartite equality question is outside its scope.

Computational and AI assistance. AI assistance was used in developing and writing the argument, producing exact Python/SymPy calculations, and conducting a separate mathematical review with a different verification implementation. That review is not external peer review or proof-assistant formalization. The explicit matrices, polynomial identities and finite certificates specify the mathematical content independently of the assistance process.

A Reconstructing the finite certificates

The supplement contains the data and two executable verifiers. From this article’s source directory, run

```
python3 supplement/verify.py
python3 supplement/verify_symbolic.py
```

The first script needs only Python’s standard library; the second requires SymPy. Both read the data without modifying it, recompute their respective certificates, and fail if a residue or polynomial coefficient disagrees.

For an unequal-endpoint row of Table 1, loop over $t = 0, \dots, p - 1$. Substitute $c = t$ into Q_3, Q_5 , form W by (11), and form the matrix in (13). To compute its determinant modulo p , choose a nonzero pivot in each column, swap rows if needed (changing the determinant sign), multiply the accumulated determinant by the pivot, and clear entries below it using the pivot inverse in $\mathbb{F}_p$. A column with no pivot yields determinant zero. These operations are exact in the finite field; the resulting determinant must match the table entry.

For the symmetric cases, one may simply evaluate the two displayed polynomials. A determinant calculation additionally verifies their elimination origin. Set

$$X = \begin{pmatrix} 0 & (2m+1)t-1 \\ 1 & t-2 \end{pmatrix}, \quad D_0 = \begin{pmatrix} 0 & 2m-3 \\ 1 & -2m \end{pmatrix}, \quad T = X \otimes I_2 + I_2 \otimes D_0.$$

These companion matrices represent multiplication by x and d in $\mathbb{F}_p[x, d]/(q_m(x), h_m(d))$. With $c = t$, compute

$$F = T^3 + (1-t)T^2 - (2mt+1)T + ((2m-3)t-1)I_4.$$

Then $P_m(t) = -\det F$. This norm identity agrees with (10); the order of the two resultant arguments in its final elimination introduces no sign, since h_m has degree two. Thus the two symmetric rows are also checked directly from their defining polynomials.

References

- [1] Y. Liu and Y. Wang, *Changes in the Seidel energy of blow-up graphs under edge deletion*, arXiv:2607.06095v1 (2026). <https://arxiv.org/abs/2607.06095v1>.
- [2] G.-X. Tian, Y. Li and S.-Y. Cui, The change of Seidel energy of tripartite Turán graph due to edge deletion, *Linear and Multilinear Algebra* **70**(19) (2022), 4597–4614. <https://doi.org/10.1080/03081087.2021.1888858>.
- [3] Z. Zeng, *Exact Verification of the Line-Graph Signature Inequality at Order Nine*, preprint (2026). <https://doi.org/10.2139/ssrn.7386040>.